BSc Degree in Applied Statistics

Statistics in Health Sciences
4. Classification of variables

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Statistics in Health Sciences

1. Classification of variables according to the measure type
2. Classification of variables according to the role in the study
3. Types of explanatory variables

Warning



During class sessions, **it is not allowed to:**

- Take photos,
- Record videos,
- Record audio.

Classification of variables according to the measure type

Quantitative variables

Objectively quantifiable:

- **Continuous**: Between two possible values of the variable, we always can find other possible value. Examples: Glucose level in blood; Exposure time to UV (ultraviolet) radiation.
- **Discrete**: Non continuous. Examples: Number of deaths; Number of hospitalizations in a given hospital.

Qualitative variables

Non objectively quantifiable:

- **Ordinal**: The possible values of the variable can be naturally ordered. Examples: Seriousness of an illness as mild, moderate or grievous; Categorised BMI (body mass index) as underweight, normal weight, overweight or obese.
- **Nominal**: Non ordinal. Examples: Profession; Marital status, and **binary or dichotomous** variables such as gender (in some cases).

Classification of variables according to the role in the study

Response or outcome

Is the variable motivating the study. Usually, it measures certain characteristic related to health, whose description or modelling is the main aim of the study. Some studies analyze several outcomes at a time.

Exposure

AKA regressor or explanatory variable, depending on the context. It is the variable that could be hypothetically related to the outcome, so its variability could (partially) explain or predict the variability of the outcome. Some studies analyze several exposure at a time, even a high number of them, as is the case of the exposome[a] paradigm.

[a] ✎ Explore "exposome" concept. See, for instance, the HELIX study at https://www.projecthelix.eu/index.php/en.

Confounder

It is a variable that could be associated with the exposure and it also could have an effect on the outcome. A confounder is not of interest itself but it should be include in the study in order to control for confounding resulting in **adjusted effects**. Usually, several confounders are considered. Typical confounders are sex and age. The concept of confusion will be introduced later in this course.

Types of explanatory variables regarding its relationship with the outcome

Covariate

- It is any potential explanatory variable included in the study.
- In some fields of biostatistics, this name is applied mainly to continuous explanatory variables.

Factor

- In general, it is any potential qualitative explanatory variable included in the study.
- In the context of epidemiological studies, if a factor has a significant negative (positive) association with the outcome, it is known as a potential risk (protection) factor.
- In R, if `x` has class *factor*, then it is considered as a qualitative variable (regardless its role in the statistical analysis).